



Constellation

10 CFR 50.73

NMP2L2880
May 1, 2024

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Nine Mile Point Nuclear Station, Unit 2
Renewed Facility Operating License No. NPF-69
Docket No. 50-410

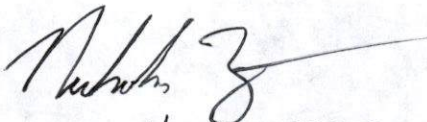
Subject: NMP2 Licensee Event Report 2024-001, Automatic Reactor Scram on Turbine Trip due to Low Condenser Vacuum

In accordance with the reporting requirements contained in 10 CFR 50.73(a)(2)(iv)(A), please find enclosed NMP2 Licensee Event Report (LER) 2024-001, Automatic Reactor Scram on Turbine Trip due to Low Condenser Vacuum.

There are no regulatory commitments contained in this letter.

Should you have any questions regarding the information in this submittal, please contact Brandon Shultz, Site Regulatory Assurance Manager, at (315) 349-7012.

Respectfully,


NICHOLAS TRYST

For Carl J. Crawford
Plant Manager, Nine Mile Point Nuclear Station

CJC/MLR

IEZZ
NRR

Enclosure: NMP2 Licensee Event Report 2024-001, Automatic Reactor Scram on Turbine Trip due to Low Condenser Vacuum

cc: NRC Regional Administrator, Region I
NRC Resident Inspector
NRC Project Manager

Enclosure

NMP2 Licensee Event Report 2024-001,
Automatic Reactor Scram on Turbine Trip due to Low Condenser Vacuum
Nine Mile Point Nuclear Station, Unit 2

Renewed Facility Operating License No. NPF-69



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Nine Mile Point Unit 2

☒ 050
☐ 052

2. Docket Number

410

3. Page

1 OF 5

4. Title

LER 2024-001-00, Automatic Reactor Scram on Turbine Trip due to Low Condenser Vacuum

5. Event Date

Month	Day	Year
03	03	2024

6. LER Number

Year	Sequential Number	Revision No.
2024	- 001 -	00

7. Report Date

Month	Day	Year
04	30	2024

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
Facility Name	☐ 052	Docket Number

9. Operating Mode

MODE 1

10. Power Level

55%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Brandon Shultz, Site Regulatory Assurance Manager

Phone Number (Include area code)

(315) 349-7012

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	TC	RBLR	M12000	Yes					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 3/3/24 at 1942 while performing a plant shutdown in preparation for a refuel outage, Nine Mile Point Unit 2 experienced a Reactor Scram due to a Main Turbine trip on low condenser vacuum. The plant was at ~ 55% power at the time of the Reactor Scram.

Following the scram, a Low RPV Level scram and containment isolation signal on Level 3 was received, as expected. The containment isolation signal impacted RHR Shutdown Cooling, RHR letdown to radwaste, and RHR sampling. All impacted valves were in the closed position at the time the isolation occurred.

All control rods were fully inserted. Plant response was as expected. Post scram the Main Turbine Bypass Valves were used to control decay heat and normal post scram level control via the feed / condensate system.

The event notification (EN 57004) was made pursuant to 10 CFR 50.72(b)(2)(iv)(B) RPS Actuation and 10 CFR 50.72(b)(3)(iv)(A) Specified System Actuation and is reportable pursuant to 10 CFR 50.73(a)(2)(iv)(A).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Nine Mile Point Unit 2	☒ 050	2. DOCKET NUMBER 410	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE

THE ENERGY INDUSTRY IDENTIFICATION SYSTEM (EIIS) COMPONENT FUNCTION IDENTIFIED AND SYSTEM NAME OF EACH COMPONENT OR SYSTEM REFERRED TO IN THIS LER ARE ENCLOSED WITHIN [BRACKETS]

I. DESCRIPTION OF EVENT**A. PRE-EVENT PLANT CONDITIONS:**

Prior to the event, NMP2 was in Mode 1 (Power Operation) at 55% power.

B. EVENT:

At 19:15 on March 3, 2024, NMP2 was continuing with a downpower in preparation for a refueling outage with normal gland sealing steam [TC] in service. As expected around 55% reactor power, the extraction steam [SB] pressure lowered to the 50 psig setpoint causing the extraction steam valve (2ESS-STV104 REBOILERS 1A & 1B EXTR SUPPLY ISOL VLV) to Clean Steam Reboiler A (CSR 'A') [RBLR] to close. By design, a main steam [SB] valve (2ASS-STV112 REBOILERS 1A & 1B MAIN STM TRIP VLV) is intended to open as soon as the extraction steam valve closes, raising pressure to 95 psig. Upon closure of 2ESS-STV104, 2ASS-STV112 failed to open automatically. Operators attempted to open the valve manually by taking the control switch to the open position, however the valve indicated on the main control board that it remained closed and steam pressure to the CSR 'A' began to lower rapidly.

To restore pressure, operators re-opened 2ESS-STV104 to re-establish the extraction steam source. CSR 'A' outlet pressure remained adequate to supply turbine seal glands, however pressure was lowering slowly consistent with extraction steam pressure. Without operator action, 2ASS-STV112 opened approximately 6 minutes later causing CSR 'A' inlet pressure to rise rapidly. Within 40 seconds, the CSR 'A' drain tank [TK] reached the high-level alarm and the hi-hi level trip was reached 6 seconds later sending a signal to isolate the inlet steam (2ASS-MOV3A REBOILER 1A STM INLET VLV) [ISV] to the CSR.

The hi-hi level cleared and recurred six additional times. During this time operators attempted to reopen 2ASS-MOV3A to CSR 'A', however it auto isolated each time. When regulated turbine gland seal pressure dropped below the 6.3 psig setpoint for the Emergency Gland Sealing [SB] steam line, the system was unavailable to support turbine glands as it had been isolated due to failures occurring during the September 2023 forced outage.

While Auxiliary Boilers [SA] were at normal pressure and temperature, valve 2ASS-AOV145 (AUX BLR STM INLET VLV) to the clean steam reboilers was gagged closed and not readily available to the operators. As a result of degrading CSR pressure and unavailability of redundant Emergency Gland Sealing steam supply, gland sealing steam pressure was lost at 19:38 and condenser [COND] vacuum rapidly degraded. Operators stopped actions to restore the CSR to brief on a manual reactor trip. Prior to initiating the manual trip an automatic trip [JC] occurred at 19:42 at an indicated condenser vacuum of 24.5".

At 19:59, CSR 'A' was restored to service by opening 2ASS-MOV3A. No additional hi-hi level conditions occurred. Operators requested maintenance support to close 2ASS-STV112 because it has an interlock that prevents 2ASS-AOV145 from being opened unless 2ASS-STV112 is closed. Once closed, the gagging device on 2ASS-AOV145 was removed and steam was supplied to the CSR from the Auxiliary boilers.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

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YEAR	SEQUENTIAL NUMBER	REV NO.					
2024	- 001	- 00					

NARRATIVE**C. INOPERABLE SYSTEMS, STRUCTURES, OR COMPONENTS THAT CONTRIBUTED TO THE EVENT:**

The Emergency Gland Sealing Steam supply was out of service.

D. DATES AND APPROXIMATE TIMES OF MAJOR OCCURENCES AND OPERATOR ACTIONS:

The dates, times, major occurrences, and operator actions for this event are as follows:

March 3, 2024:

- 19:15 2ESS-PS112 setpoint reached (< 50 psig Extraction Steam) at 55% reactor power
2ESS-STV104 closes in auto as expected
Operators recognize that 2ASS-STV112 did not open in auto
- 19:16 Operators attempt to manually open 2ASS-STV112 by taking the control switch to the open position after it fails to open in auto
Valve position does not appear to change
- 19:17 2ESS-STV104 Re-opened by taking the control switch out of the auto position and to the open position
Clean Steam Reboiler Pressure discharge pressure recovers from 27 psig to 35 psig
- 19:23 2ASS-STV112 Opens unexpectedly
Clean Steam Reboiler Drain Tank Hi-Hi Level, resulting in CSR isolation.
- 19:25 Clean Steam Reboiler Outlet Pressure Peaks and then lowers consistently until turbine trip
- 19:29 Clean Steam Reboiler Drain Tank Hi-Hi Level occurs a total of 7 times between 19:23 and 19:42, with at least three of these occurring as operators attempt to manually open the CSR steam admission valve 2ASS-MOV3A
- 19:38 Regulated turbine gland sealing steam pressure 0 psig
- 19:42 Turbine Trip on Low Condenser Vacuum and Reactor Scram
- 19:59 Operators successfully open 2ASS-MOV3A and Gland Sealing Steam Pressure recovers

E. METHOD OF DISCOVERY:

This event was self-revealed when the turbine tripped on Low Condenser Vacuum

F. SAFETY SYSTEM RESPONSES:

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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2024	- 001	- 00					

NARRATIVE

All safety systems responded as designed.

II. CAUSE OF THE EVENT

The direct cause of the event was the isolation of Clean Steam Reboiler 'A.' The loss of gland sealing steam and unavailability of the Emergency Gland Sealing steam line resulted in degrading condenser vacuum and turbine trip. The isolation was caused by the CSR 'A' drain tank level control system, which was unable to maintain level below the Hi-Hi isolation signal. Deficiencies were identified with the level control system and drain flow paths, resulting in the inability of the system to respond to Hi and Hi-Hi level conditions rapidly.

III. ANALYSIS OF THE EVENT

All safety systems responded per design. No principal barriers were degraded or lost. Gland sealing steam was returned to service via the Auxiliary Boilers shortly after the scram, and decay heat removal occurred via the main condenser. The scram did not have impact to the health or safety of the public.

Subsequent investigation of the system identified multiple degraded components which, in the aggregate, resulted in a loss of the gland sealing steam system. The unavailability of the Emergency Gland Sealing steam supply resulted in loss of redundancy. This system was isolated due to failure in September 2023 and was unavailable to perform its function on March 3, 2024.

The normal and dump level control systems for the CSR 'A' drain tank both exhibited partial losses of function to respond quickly to level increases. The normal level control solenoid 2CNA-SOV101 was found not transmitting full control air signal to level controller 2CNA-LIC101. As a result, the normal drain system would only respond partially to high level conditions. The dump valve level transmitter 2CNA-LT9A was found degraded with the inability to begin throttling open the dump valve at the expected setpoint. In this degraded condition the valve would not modulate as expected. The combined degraded conditions of the drain systems resulted in isolation of the CSR and failure to maintain gland sealing steam pressure to the turbine.

The CSR and Emergency Gland Sealing Steam supply systems combined have resulted in two prior scrams (2006, 2012). In these prior events, different components were the direct cause of the scram.

IV. CORRECTIVE ACTIONS**A. ACTION TAKEN TO RETURN AFFECTED SYSTEMS TO PRE-EVENT NORMAL STATUS:**

The station completed repairs to the Emergency Gland Sealing steam system, replaced 2CNA-SOV101, replaced 2CNA-LIC101, and repaired 2CNA-LT9A.

B. ACTION TAKEN OR PLANNED TO PREVENT OCCURRENCE:

The station is implementing a change to the system management strategy to ensure aggregate conditions do not preclude satisfactory system operation. A design modification is being developed for later implementation.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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NARRATIVE

V. ADDITIONAL INFORMATION

A. FAILED COMPONENTS:

Clean Steam Reboiler 'A'

B. PREVIOUS LERs on SIMILAR EVENTS:

NMP Unit 2 2006-001-00, Automatic Reactor Scram due to a Loss of Main Turbine Gland Sealing Steam

NMP Unit 2 2012-004-00, Manual Reactor Scram due to a Loss of Main Turbine Gland Sealing Steam
Resulting in Lowering Condenser Vacuum